

Optimal Transport and Sinkhorn's Algorithm

An Expository Overview

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AMSC808 project presentation

May 7, 2026

1 Introduction

The Optimal transport problem (with its roots going back to Monge in 1781), is described as follows:

(Static) OT Problem: *Given two probability measures (Radon) α and β defined on measure spaces X and Y , and a cost function $c : X \times Y \rightarrow \mathbb{R}$, find a measurable map $T^* : X \rightarrow Y$ that minimizes the total transport cost:*

$$T^* = \operatorname{argmin}_T \left\{ \int_X c(x, T(x)) d\alpha(x) \mid T_{\#}\alpha = \beta \right\} \quad (1)$$

where $T_{\#}\alpha$ denotes the pushforward measure.

In discrete spaces of equal cardinality with uniform measures, this problem reduces to the Optimal Assignment Problem (popularized in the 1950s):

Otpimal Assignment Problem: *Given a cost matrix $(C_{i,j})$, we seek a permutation σ that minimizes $\sum_{i=1}^n C_{i,\sigma(i)}$.*

A fundamental challenge of the Monge problem and the optimal assignment problem is that they are highly non-convex, and in many settings, a Monge map T^* might not even exist.

2 The Kantorovich Relaxation

2.1 Mass-splitting vs Mass-conservation

To address the non-existence of Monge maps, a probabilistic, convex relaxation was introduced. Instead of strict mass conservation via a deterministic Monge map T^* , the Kantorovich formulation allows for mass splitting in the following sense: We define the set of valid couplings $\mathcal{U}(\alpha, \beta)$ as the space of joint probability measures $\pi \in \mathcal{M}_+^1(X \times Y)$ whose marginals are α and β respectively. The relaxed problem becomes finding:

$$\pi^* = \operatorname{argmin}_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y) \quad (2)$$

It is easy to see to get a sense of why this is a relaxation of the original problem. In the discrete case, where $\alpha = \sum_{i=1}^n a_i \delta_{x_i}$ and $\beta = \sum_{j=1}^m b_j \delta_{y_j}$, the Monge problem seeks a map

$T^* : \{x_1, \dots, x_n\} \rightarrow \{y_1, \dots, y_m\}$ such that

$$\forall j \in \{1, \dots, m\}, \quad b_j = \sum_{i: T(x_i)=y_j} a_i,$$

while the Kantorovich problem seeks a coupling matrix $P \in \mathbb{R}_+^{n \times m}$ such that $P\mathbf{1}_m = \vec{a}$ and $P^\top \mathbf{1}_n = \vec{b}$. In this latter (now convex) problem, P_{ij} denotes the mass of a_i that is transported to b_j (thus the term mass splitting). Indeed, we also have that for the general Monge map T^* , the measure $\pi = (I \times T)_\# \alpha$ acts as a solution to this relaxation. In fact, it can also be checked that this relaxation is tight for the Optimal Assignment Problem, where such an optimal coupling is necessarily a permutation. We also have the following well-posedness theorem for the Kantorovich problem.

Well-posedness of KP: *When X and Y are Polish spaces and the cost c is lower semi-continuous, a solution to the Kantorovich relaxation is guaranteed to exist via the direct method of calculus of variations*

Sketch of Proof. X, Y being Polish is enough to imply that $U(\alpha, \beta)$ is weakly compact, while c being l.s.c is enough to imply that the functional $I[\pi] = \int_{X \times Y} c(x, y) d\pi(x, y)$ is weakly l.s.c. $\square$

2.2 p -Wasserstein Metrics

When $X = Y$ and the cost is the p -th power of a metric distance, $c(x, y) = d(x, y)^p$ (for $p \geq 1$), the minimal cost induces a metric on the space of probability measures (with finite p -moments) known as the p -Wasserstein distance:

$$\mathcal{W}_p(\alpha, \beta) = \left(\inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{X \times Y} d(x, y)^p d\pi(x, y) \right)^{1/p} \quad (3)$$

For $X = \mathbb{R}^d$ and $c(x, y) = \|x - y\|^2$, we have

$$W_2(\alpha, \beta)^2 = W_2(\tilde{\alpha}, \tilde{\beta})^2 + \|m_\alpha - m_\beta\|^2,$$

where

$$\tilde{\alpha} = (T_{m_\alpha})_\# \alpha, \quad \tilde{\beta} = (T_{m_\beta})_\# \beta.$$

$T_\tau : x \mapsto x - \tau$, and m_α, m_β are the means of α, β respectively. And so, the $\mathcal{W}_p$ distance between measures accounts for spatial shifts, providing an intuitive distance metric even when measures do not have overlapping supports.

2.3 The Dual Problem and Brenier's Theorem

The strong dual to the Kantorovich problem allows us to reframe the computation using Kantorovich potentials, φ and ψ :

$$\sup_{\substack{\varphi \in C_b(X) \\ \psi \in C_b(Y)}} \left\{ \int_X \varphi(x) d\alpha(x) + \int_Y \psi(y) d\beta(y) \mid \varphi(x) + \psi(y) \leq c(x, y) \right\} \quad (4)$$

Proof. Introduce the marginal constraint in (2) as a penalty to form the Lagrangian

$$\mathcal{L}(\pi, \varphi, \psi) = \int c(x, y) d\pi(x, y) + \left(\int \varphi(x) d\alpha(x) - \int \varphi(x) d\pi(x, y) \right) + \left(\int \psi(y) d\beta(y) - \int \psi(y) d\pi(x, y) \right).$$

Note that we must have $\inf_{\pi \in \mathcal{M}_+(X \times Y)} \sup_{\substack{\varphi \in C_b(X) \\ \psi \in C_b(Y)}} \mathcal{L}(\pi, \varphi, \psi) = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int c(x, y) d\pi(x, y)$.

This is because if $\pi \notin U(\alpha, \beta)$, i.e, the marginal condition is not satisfied, the penalty terms can

be blown up to infinity by scaling ψ and φ . Now using *Sion's Minimax Theorem* (under the appropriate conditions of c being l.s.c and X, Y being Polish), we may switch the inf and sup. Evaluating the inner infimum now, we can note that if $\varphi(x) + \psi(y) > c(x, y)$ at some (x, y) , we can increasingly concentrate the measure $\pi \in \mathcal{M}_+(X \times Y)$ such that the first term in the Lagrangian

$$\mathcal{L}(\pi, \varphi, \psi) = \int [c(x, y) - \varphi(x) - \psi(y)] d\pi(x, y) + \int \varphi(x) d\alpha(x) + \int \psi(y) d\beta(y)$$

can be made arbitrarily negative. Thus we obtain the constraint in (4). $\square$

This leads to a landmark result bridging the Monge and Kantorovich formulations.

Brenier's Theorem (1987) *For $X = Y = \mathbb{R}^d$ and $c(x, y) = \|x - y\|^2$, α absolutely continuous with respect to the Lebesgue measure, and α, β having finite second moments:*

1. *The optimal coupling π^* in KP is unique.*
2. *π^* is supported on the graph of a Monge map T^* , i.e., $\pi^* = (I \times T^*)_{\#}\alpha$.*
3. *T^* is uniquely defined as the gradient of a convex function f , i.e., $T^*(x) = \nabla f(x)$, where f is the unique (up to an additive constant) convex function such that $f(x) = \frac{1}{2}\|x\|^2 - \varphi(x)$.*

Note: Under extra regularity assumptions, this leads to another equivalent formulation given by the Monge-Ampère equation:

$$\det(\nabla^2 f(x)) \rho_{\beta}(\nabla f(x)) = \rho_{\alpha}(x)$$

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3 Discrete Problem

3.1 Entropic Regularization

In the setting where the measures are discreetly supported and defined by probability vectors a, b , the problem is constrained to $U(a, b) = \{P \in \mathbb{R}_+^{m \times n} \mid P\mathbf{1}_n = a, P^\top \mathbf{1}_m = b\}$ and

$$P^* = \operatorname{argmin}_{P \in U(a, b)} C : P,$$

which is a standard linear program (and hence not necessarily unique solutions). However, scaling this up computationally is expensive. To resolve this, one can introduce **Entropic Regularization**. By adding an entropy penalty, the problem becomes strictly convex:

$$P_{\epsilon} = \operatorname{argmin}_{P \in U(a, b)} C : P - \epsilon H(P) \tag{5}$$

where $H(P) = -\sum_{i,j} P_{i,j}(\log P_{i,j} - 1)$. The strong convexity guarantees a unique solution P_{ϵ} over the convex and compact feasible set $U(a, b)$. As $\epsilon \rightarrow 0$, P_{ϵ} provably converges to the unique maximum-entropy solution of the unregularized problem, i.e

$$P_{\epsilon} \rightarrow \operatorname{argmax}\{H(P) : P \in \operatorname{argmin}_{P' \in U(a, b)} C : P'\}.$$

Note that this is unique since the solution set of the unregularized problem is convex.

Proof. Define $\hat{U} := \{P \in (\mathbb{R}_+)^{m \times n} : \mathbf{1}_m \mathbf{1}_n^\top : P = 1\} \supset U(a, b)$. Since $\hat{U}$ is convex and compact, $H(P)$ attains a unique maxima on it. By symmetry of the entropy and the constraint defining

$\hat{U}$, this must occur at $P = \frac{\mathbf{1}_m \mathbf{1}_n^T}{mn}$ with value $1 + \log(mn)$. Let $P^* \in \operatorname{argmin}_{P \in U(a,b)} C : P$. Then by definition

$$\begin{aligned} C : P^* &\leq C : P_\epsilon \\ C : P_\epsilon - \epsilon H(P_\epsilon) &\leq C : P^* - \epsilon H(P^*), \end{aligned}$$

and so

$$0 \leq C : P_\epsilon - C : P^* \leq \epsilon(H(P_\epsilon) - H(P^*)) \leq \epsilon(1 + \log(mn)). \quad (6)$$

Note also that $H(P_\epsilon) \geq H(P^*)$. By compactness of $U(a,b)$, P_ϵ converges upto a subsequence to some $P_0 \in U(a,b)$ with $H(P_0) \geq H(P^*)$. Hence $P_0 \in \operatorname{argmin}_{P \in U(a,b)} C : P$ by (6) and $P_0 = \operatorname{argmax} \{H(P) : P \in \operatorname{argmin}_{P' \in U(a,b)} C : P'\} \Rightarrow P_\epsilon \rightarrow P_0$ without a subsequence. $\square$

3.2 The Sinkhorn Algorithm

By writing the Lagrangian of the regularized problem and differentiating, we find the optimality condition:

$$P_{ij} = e^{-C_{ij}/\epsilon} e^{f_i/\epsilon} e^{g_j/\epsilon} \quad (7)$$

This implies that the optimal matrix P takes the form $P = \operatorname{diag}(u)K\operatorname{diag}(v)$, where $K_{ij} = e^{-C_{ij}/\epsilon}$, and u, v are positive vectors. Thus, it can be shown that solving for P amounts to the solving a matrix-scaling problem

$$\begin{aligned} u \odot (Kv) &= a \\ v \odot (K^T u) &= b, \end{aligned}$$

This is numerically solved via **Sinkhorn's algorithm (1964)**. The algorithm proceeds via alternating updates:

$$u^{l+1} = \frac{a}{Kv^l}, \quad v^{l+1} = \frac{b}{K^T u^l} \quad (8)$$

where division is element-wise. Recent error analysis shows that setting $\epsilon = \frac{\tau}{4\log(n)}$, it takes $O(\|C\|_\infty \log(n) \tau^{-3})$ iterations to achieve $C : \hat{P} \leq \min_{P \in U(a,b)} C : P + \tau$, where $n = m$ assumed for simplicity [1]. Thus the primary computational bottleneck is the dense matrix-vector product with K . We discuss an example of computing this next.

3.3 Distance as Cost

Assume $c(x, y) = d_{\mathcal{M}(x,y)}$ where $\mathcal{M}$ is a Riemannian manifold with some inverse metric tensor $M(x)$. Let $\Delta_{\mathcal{M}} v(x) = M(x) : \nabla^2 v(x)$ be the *Laplace-Beltrami* operator. Assuming $\mathcal{M}$ has parametric coordinates in $y \in \Omega \subset \mathbb{R}^d$, it can be shown that $u_y = d_{\mathcal{M}}(\cdot, y)$ is a solution to the *eikonal equation*

$$\begin{aligned} \nabla u_y \cdot M(x) \nabla u_y - 1 &= 0 \quad x \in \Omega \setminus \{y\} \\ u_y(y) &= 0 \end{aligned}$$

and that $v_y^\epsilon = \exp(-u_y/\epsilon)$ approximately solves

$$\begin{aligned} \epsilon^2 \Delta_{\mathcal{M}} v_y^\epsilon - v_y^\epsilon &= 0 \quad x \in \Omega \setminus \{y\} \\ v_y^\epsilon|_{\partial\Omega} &= 0, \quad v_y^\epsilon(y) = 1 \end{aligned}$$

Numerical PDE methods can be used to get K . This is usually FEM in the case of general $\mathcal{M}$, but in the euclidean case, FDM can also apply.

References

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